Grounded Two-Way Visual Dialogue: Interactive Multimodal Reasoning, Cross-View Geometric Verification, and Calibrated Perception

PhD Project Description & Research Proposal

Candidate: Silas Kwabla Gah, MPhil in Computer Science (University of Ghana)

Email: gahsilas@gmail.com | Target: Visual Computing Section, DTU Compute, Technical University of Denmark

1. Vision: Beyond Text-Only Interaction to Two-Way Visual Dialogue

Multimodal Large Language Models (MLLMs) demonstrate strong conversational fluency, yet practical human-AI collaboration requires moving beyond text-only modalities into bidirectional visual communication. In physical environments such as robotics, collaborative design, augmented reality, and medical imaging, interaction naturally relies on visual references, pointing, spatial boundaries, and reciprocal visual feedback.

A practical visual dialogue system requires three core capabilities:

1. **Input Comprehension:** Accurately parsing dense, ambiguous human visual cues (e.g., natural descriptions, bounding regions, and multi-view camera perspectives).
2. **Reciprocal Visual Output:** Responding with grounded spatial elements such as dynamic segmentation masks, 3D bounding boxes, and clarifying visual queries when input instructions are ambiguous.
3. **Calibrated Spatial Uncertainty:** Providing honest confidence estimates regarding spatial predictions, initiating clarification steps rather than generating ungrounded locations.

This proposal outlines a 3-year research plan within the Visual Computing Section at DTU Compute to develop and evaluate architectures for grounded two-way visual dialogue.

2. Methodological Foundation and Prior Research

My prior work provides relevant methodological foundations for this research:

- **Geometric Process Supervision (*Gah et al., SACAIR 2026*):** Supervised multimodal spatial reasoning in Qwen2.5-VL by projecting 2D open-vocabulary predictions (SAM3) into shared 3D coordinates (RegionPLC, OpenScene). Using physical cross-view consistency as an intrinsic reward, we trained policies with online Group Relative Policy Optimization (GRPO) in VERL.
- **Addressing Specification Gaming in Spatial Grounding:** Identified an enumeration failure mode where spatial policies gamed recall by predicting oversized regions. We resolved this by formulating count-based precision and weighted $F_{0.5}$ metrics, reducing area bias correlation from $\rho > 0.72$ to 0.334.
- **Reliability-Calibrated Adaptive Semantic Arbitration (RCASA) (*IEEE TPAMI Under Review*):** Formulated a probabilistic framework using Platt confidence calibration and Bernoulli uncertainty estimation to coordinate frozen foundation models at inference time ($\Delta\theta = 0$), achieving competitive results on PASCAL-5^i (77.9% HM) and ScanNet200 (33.68% HM).

3. Proposed 3-Year Research Plan (DTU Compute)

Year 1: Multi-Turn Grounding and Disambiguation Datasets

- **Objective:** Construct multi-turn visual dialogue benchmarks capturing sequential human-AI disambiguation (e.g., user pointing $\rightarrow$ model proposes candidate masks with confidence metrics $\rightarrow$ user refines $\rightarrow$ model confirms).
- **Methodology:** Combine multi-view 3D environments (ScanNet++, Habitat) with interactive

collection protocols to curate diverse spatial reference trajectories.

- **Milestone 1:** Conference submission to CVPR / ICCV on grounded visual dialogue benchmarks.

Year 2: End-to-End Two-Way Visual Dialogue Architectures with RLVR

- **Objective:** Develop MLLM architectures that natively output interleaved text and spatial tokens (masks, 3D boxes, coordinates) trained via Reinforcement Learning with Verifiable Rewards (RLVR).
- **Methodology:** Extend cross-view geometric consistency rewards to dynamic conversational turns, enforcing spatial-temporal consistency across successive dialogue states.
- **Milestone 2:** Publication at ECCV / NeurIPS on grounded conversational vision architectures.

Year 3: Uncertainty Calibration, User Studies, and International Stay

- **Objective:** Integrate RCASA probabilistic calibration to enable uncertainty-aware visual dialogue and complete the required DTU international research exchange.
- **Methodology:** Evaluate models in interactive human-AI user studies to measure task completion, precision, and interaction latency. Complete PhD thesis preparation and defense.
- **Milestone 3:** Journal manuscript submission (*IEEE TPAMI / IJCV*) and PhD dissertation defense.

4. Synergy with DTU Compute and Teaching Activities

- **Research Alignment:** DTU Compute's Visual Computing section provides an established environment for geometric computer vision, 3D scene understanding, and interactive AI.
- **Teaching Participation:** Prepared to serve as a Teaching Assistant for DTU courses in Computer Vision, Deep Learning, and Machine Learning, supported by 4+ years of prior university lecturing experience.